

Longitudinal Drug Response Assessment of Tumor Organoids Based on Optical Attenuation Coefficient and Multi-dimensional Morphological Characterization

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Abstract: Patient-derived tumor organoids (PDTOs) are vital for precision oncology, but standard drug screening methods, including adenosine triphosphate (ATP) assays, are destructive and prevent longitudinal monitoring. Optical coherence tomography (OCT) offers non-destructive 3D imaging, capturing morphological features and tissue attenuation characteristics via the optical attenuation coefficient (OAC). Here, we propose a non-destructive evaluation framework for tumor organoids fusing OAC and multi-dimensional morphological features. Using intrahepatic cholangiocarcinoma (iCCA) PDTOs treated with icaritin, we found that OAC exhibited a significant dose-dependent increase (up to 32.8% at 80 μ M compared to control), accompanied by a morphological transition of cystic organoids into solid phenotypes. By integrating these features via K-means++ clustering and Principal Component Analysis, we constructed a relative growth score. This fusion score correlated strongly with the ATP gold standard (Pearson correlation coefficient $r=0.938$), outperforming a morphology-only model ($r=0.906$). Furthermore, independent experiments with first-line chemotherapeutics (e.g., 5-Fluorouracil, Gemcitabine) and combinatorial regimens indicated the model's potential generalizability ($r=0.887$). This method overcomes the limitations of destructive, single-metric evaluations, providing a quantitative and non-destructive platform for high-throughput drug screening and personalized treatment decision-making.

Keywords: tumor organoids; drug screening; optical attenuation coefficient; drug response assessment; morphological characterization

1. Introduction

Tumor heterogeneity leads to individual variations in therapeutic efficacy and clinical drug resistance. Therefore, in vitro models capable of precisely predicting drug responses are crucial breakthroughs for realizing precision medicine^[1-3]. Patient-derived tumor organoids (PDTOs), as tissue-engineered models formed through three-dimensional culture in biomaterial scaffolds (e.g., Matrigel), can accurately preserve the histological features, genetic mutation profiles, and functional heterogeneity of

parental tumors, and are widely recognized as ideal in vitro models bridging basic research and clinical treatment ^[4-7]. Currently, PDOs have been widely applied in high-throughput drug screening, research on drug resistance mechanisms, and the formulation of personalized treatment regimens ^[8-10].

Despite these advantages, the dynamic and non-destructive quantification of PDO drug responses remains a major technical challenge. Existing gold-standard methods, such as the measurement of adenosine triphosphate (ATP) levels or mitochondrial reduction capacity (e.g., MTT/CCK-8 assays), are destructive endpoint tests; they not only fail to support the longitudinal dynamic monitoring of precious patient samples but also yield results that are easily confounded by inter-well initial growth variations ^[11, 12]. Among conventional imaging technologies, although bright-field microscopy enables label-free observation, it lacks three-dimensional resolution capabilities and provides limited contrast for transparent organoids. Confocal microscopy and fluorescence imaging can provide high-resolution three-dimensional structural information, but they typically rely on exogenous fluorescent labeling, which poses risks of photobleaching and phototoxicity, making it difficult to support the long-term, non-destructive dynamic tracking of organoids. Therefore, a label-free, three-dimensional, and long-term compatible imaging modality is urgently required to more authentically reveal the dynamic evolutionary process of organoids under drug treatment ^[13-15].

Optical Coherence Tomography (OCT), leveraging its non-destructive and three-dimensional imaging capabilities, has been applied to the long-term culture monitoring of organoids, with its analytical dimensions encompassing various aspects from macroscopic morphology (e.g., volume, surface area) to dynamic activity ^[16-22]. However, existing drug response evaluations for PDOs predominantly focus on the quantified analysis of macroscopic geometric morphological parameters, such as area, perimeter, and morpho phenotypic classification ^[23, 24]. During the early stages of drug action, or under specific mechanisms such as induced cell cycle arrest, the macroscopic volume of organoids may not have exhibited significant alterations, yet their internal microenvironment (e.g., cell density, nucleocytoplasmic ratio) has undergone drastic changes. This morphology-function temporal discrepancy can easily lead to false-negative evaluations ^[13, 25].

To overcome the aforementioned limitations, this study employs optical attenuation coefficient (OAC) derived from OCT signals. The OAC quantifies the attenuation of light within tissues, primarily originating from the scattering effect, and is closely associated with the size, density, and refractive index distribution of intracellular sub-cellular structures (e.g., nuclei, organelles) ^[26, 27]. Processes such as drug-induced apoptosis are accompanied by pronounced microscopic-scale changes, including pyknosis, chromatin condensation, and organelle remodeling. These alterations significantly change the optical attenuation characteristics of local tissues, allowing for the non-destructive detection of dead-cell and necrotic regions ^[28], and typically precede macroscopic morphological changes ^[29-31]. Moreover, OAC variations are correlated with specific pathophysiological alterations, such as gliosis following nervous system injury

[32]. This demonstrated that OAC possesses the potential to serve as a sensitive endogenous biomarker for the early warning of drug responses in organoids.

Based on this, the present study proposes a non-destructive, dynamic evaluation framework for tumor organoid drug responses, fusing OAC with multi-dimensional morphological features. Utilizing PDOs of intrahepatic cholangiocarcinoma (iCCA) as a model, the precise three-dimensional segmentation of organoids was first achieved through a deep learning algorithm to extract multi-dimensional morphological parameters such as volume and surface area. Subsequently, an optimized depth-resolved algorithm was employed to extract the spatially resolved OAC from OCT signals. To address the inherent heterogeneity of organoid populations, unsupervised clustering was introduced to classify organoid phenotypes. Combined with Principal Component Analysis (PCA), a comprehensive growth level model and a relative growth level model fusing organoid attenuation characteristics and macroscopic morphology were constructed. This method provides a novel quantitative analytical tool featuring both high information density and non-destructive characteristics for high-throughput drug screening and personalized drug sensitivity testing based on PDOs.

2. Materials and methods

2.1. Overall experimental workflow

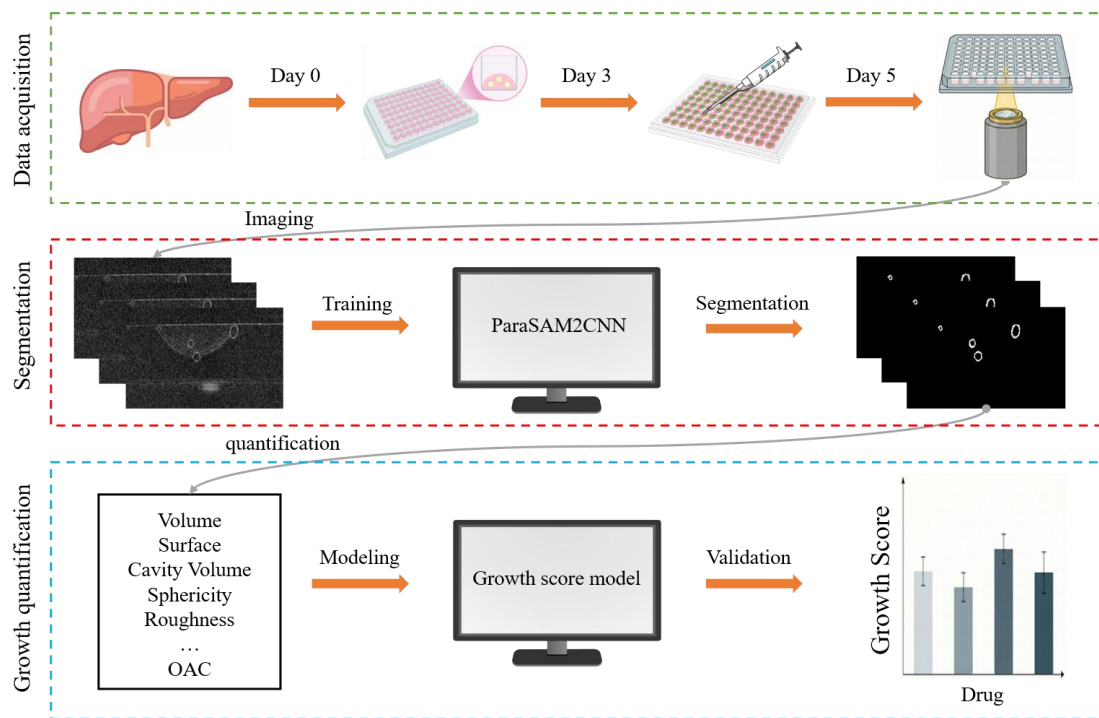


Fig. 1. Organoid collection and data processing process

The overall workflow of this study, encompassing the establishment of patient-derived iCCA organoids, standardized culture, drug treatment, and multi-modal data analysis, is illustrated in Fig. 1. Experimental samples were obtained from surgically resected iCCA tissues. This study was approved by the Institutional Research Ethics Committee of the First Affiliated Hospital, College of Medicine, Zhejiang University (approval no. 2019-826), and all procedures were conducted in accordance with the

Declaration of Helsinki, with written informed consent obtained from each subject. Following removal of necrotic tissues and mechanical mincing, tumor cell clusters were isolated via collagenase II digestion. The resulting cell pellets were mixed with Matrigel (Corning, USA) at a 2:1 volume ratio (Matrigel to cell suspension) and seeded as dome structures into 96-well plates (5 μ L/well). After Matrigel polymerization, an optimized Advanced DMEM/F12-based complete medium supplemented with essential growth factors, including EGF and R-spondin 1, was added. Organoid cultures were subsequently maintained in a humidified incubator at 37°C with 5% CO₂.

Icaritin is a prenylated flavonoid derived from *Epimedium* species. It received conditional approval from the National Medical Products Administration of China for the treatment of advanced hepatocellular carcinoma in 2022 ^[33]. A real-world study reported disease control rates of 60.0% for icaritin monotherapy and 93.8% for combination therapy ^[34], and emerging evidence suggests its potential efficacy against iCCA through immunomodulatory mechanisms ^[35]. Therefore, icaritin was selected as the model drug in this study to evaluate the treatment response of iCCA organoids. Stock solutions were prepared using dimethyl sulfoxide (DMSO). The experiments included a DMSO blank control group and three drug concentration gradient groups (20 μ M, 40 μ M, and 80 μ M). As illustrated in Fig. 1, a detailed schedule of time points was designed: following seeding on Day 0, organoids were cultured statically until Day 3. On Day 3, the initial OCT scan was performed to acquire pre-treatment morphological and optical characteristics; subsequently, the old medium was removed and replaced with fresh medium containing preset concentrations of icaritin to initiate drug intervention. On Day 5 (48 hours post-treatment), a second OCT imaging session was conducted. Immediately afterward, ATP bioluminescence detection was performed to determine cell metabolic activity as gold-standard validation data.

During data analysis, multiple feature parameters of individual organoids within the organoid population were first extracted based on OCT image segmentation. Subsequently, cluster analysis of organoids was performed using the feature parameters from the two time points (first dimensionality reduction). Statistical analysis of the clustering results yielded quantitative information for each cluster, including organoid count, mean and variance of each feature parameter. On this basis, principal component analysis was conducted to construct a growth level model (second dimensionality reduction). Next, the growth level scores at the endpoint and the differences in growth level scores between the two time points were used to characterize growth status (third dimensionality reduction). Finally, the growth level scores and relative growth scores were compared with the organoid viability indicator (ATP content) to evaluate the drug effect on the organoid population.

2.2. OCT imaging system and data acquisition

A custom-built high-resolution swept-source optical coherence tomography (SS-OCT) system was employed in this study ^[36]. The system employs a swept-source with a central wavelength of 1310 nm and a line scan rate of up to 50 kHz. The light source has a full width at half maximum of 105 nm. The axial resolution of the system in air

is 7.5 μm , and the lateral resolution is approximately 15.6 μm . The data acquisition field of view is 4 mm (x) $\times$ 4 mm (y) $\times$ 5.6 mm (z), and the acquired OCT volumetric data size is 800 (x) $\times$ 800 (y) $\times$ 1024 (z) pixels, with an image pixel resolution of 5 μm (x) $\times$ 5 μm (y) $\times$ 5.5 μm (z). To meet the requirements of high-throughput, multi-time-point detection of organoids in standard 96-well plates, an optimized inverted imaging scheme was adopted: the OCT probe was mounted upside down on a high-precision 3D translation stage while the culture plate was fixed above. This configuration avoids potential vibration interference and liquid disturbances caused by moving the culture plate. In addition, a low-numerical-aperture scan lens with a focal length of 36 mm was chosen to ensure that the organoid samples remained within the lens depth of focus. The entire imaging process was conducted in a customized 37 $^{\circ}\text{C}$ constant-temperature environment.

2.3. Image preprocessing and deep learning-based segmentation

Original OCT interference signals were reconstructed into 3D structural images after k-domain linearization, dispersion compensation, and fast fourier transform (FFT). A 3 $\times$ 3 $\times$ 3 median filtering was applied to suppress speckle noise, enhancing the signal-to-noise ratio while preserving edge details. Automated segmentation was performed using the ParaSAM2CNN deep learning model previously proposed by our team [37]. This model employs a parallel encoder architecture: one branch utilizes ResNet to extract deep semantic features, while the other branch captures multi-scale contextual information using adapters from SAM2 (Segment Anything Model 2), effectively addressing issues of blurred organoid boundaries and discontinuous 3D contours. For the initial segmentation results, the equivalent spherical diameter of each connected component was calculated, and only components with an equivalent diameter greater than 50 μm were retained to filter out small objects potentially generated by noise or cell debris, ensuring that subsequent analysis focused on well-developed organoids.

2.4. Multi-dimensional feature extraction

Based on the 3D masks obtained from segmentation and the original OCT signals, the system extracted two types of features to comprehensively quantify the morphological and optical properties of the organoids, obtaining a multi-dimensional feature set covering macroscopic morphology, surface texture, and attenuation characteristics.

1) Morphological features: These include organoid volume, fill volume, cavity volume, surface area, cavity number, cyst thickness, longest axis, shortest axis, and sphericity. Surface roughness was calculated using an adaptive power spectrum filtering algorithm [37], characterizing surface shrinkage or topological complexity changes potentially induced by drug action by quantifying the root-mean-square deviation of the surface point cloud relative to the best-fit surface.

2) Optical attenuation coefficient: Based on the single-scattering model and the Beer-Lambert law, and assuming that light attenuation within biological tissues is dominated by scattering and is locally isotropic, the OAC was obtained at the single-voxel level [26, 27]. An optimized depth-resolved OAC algorithm [38] was employed, incorporating corrections for the confocal effect and sensitivity roll-off:

$$\mu_s[i] = \frac{I[i]/A[i]}{2 \cdot \Delta \cdot \sum_{j=i+1}^N I[j]/A[j]}, \quad (1)$$

where $\mu_s[i]$ represents the OAC at the i -th pixel along the axial direction; $I[i]$ denotes the OCT signal at the i -th pixel; $A[i]$ is the correction coefficient at the i -th pixel; Δ is the pixel resolution (5.5 μm); and N represents the number of axial pixels. In addition to the spatially resolved OAC, it was correlated with the organoid segmentation results to calculate the mean (OAC Mean) and standard deviation (OAC Std) of μ_s for both the entity and fill regions, serving as quantitative optical metrics reflecting the tissue density and physiological state of individual organoids.

2.5. K-means++-based clustering of organoid types

Based on the aforementioned multi-dimensional morphological characterization parameters and OAC metrics (as shown in Fig. 2), we performed K-means++ clustering analysis using data from a total of 23,439 organoids across 24 wells at two time points to construct an organoid classification model. First, a multi-dimensional feature matrix was constructed, encompassing geometric morphology (e.g., volume, sphericity, cavity volume) and attenuation characteristics (mean OAC and OAC standard deviation). To eliminate scale discrepancies caused by different dimensions (e.g., volume in μm^3 versus dimensionless sphericity), all input features were subjected to Z-score standardization to achieve a mean of 0 and a standard deviation of 1. This ensured that each feature contributed equally to the clustering distance weights.

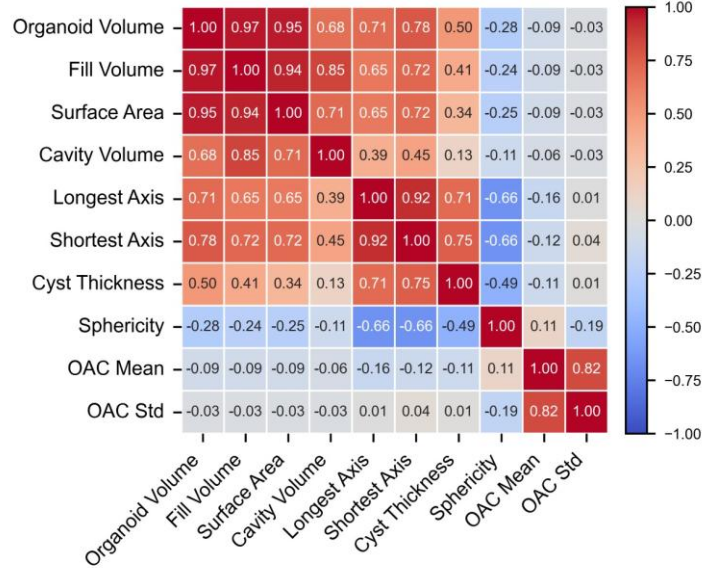


Fig. 2. Correlation heatmap of characterization features

To determine the optimal number of clusters that best fit the intrinsic data structure, the elbow method was employed. By calculating the sum of squared errors (SSE) within organoid clusters for varying k values, the k value at which the SSE descent rate showed a distinct inflection point was selected as the optimal cluster number. To prevent the algorithm from converging to local optima, the K-means++ initialization strategy was used to select the initial centroids. The maximum number of iterations was set to 300,

and the algorithm was run with 10 different initialization seeds to select the best convergence result. Finally, this clustering model was utilized to obtain category information for each individual organoid, facilitating statistical analysis based on organoid classes.

2.6. Construction of the comprehensive evaluation model

Each individual organoid was characterized by 10 morphological and attenuation features (e.g., volume, sphericity, cavity volume, mean OAC, OAC standard deviation, etc.). Based on the K-means++ clustering results, the organoids within each organoid cluster were classified into four distinct classes. For each class, we calculated the mean values of the 10 features along with the number of organoids belonging to that class, yielding 11 class-based statistical features per class. Thus, each organoid cluster was represented by 44 features (achieving the first level of dimensionality reduction).

Using the icaritin-treated experimental groups described in Section 2.1 (including a blank control group and treatment groups at 20 μM , 40 μM , and 80 μM , totaling 24 wells) as modeling samples, we constructed an initial feature matrix of dimensions 24 (clusters) $\times$ 44 (features). To reduce redundancy, we performed feature selection based on Pearson correlation analysis combined with a variance threshold method, ultimately retaining 15 category-based statistical characterization parameters. The resulting feature matrix thus had dimensions of 24 $\times$ 15.

Subsequently, PCA was applied to this 24 $\times$ 15 matrix. Through orthogonal transformation, the original correlated variables were converted into a set of linearly independent principal components (achieving the second level of dimensionality reduction). Following the Kaiser criterion, only principal components with eigenvalues $\lambda > 1$ were retained, thereby reducing the dimensionality of the data while preserving the primary variance information of the original dataset.

To compress the multi-dimensional information into a single quantitative metric, an organoid comprehensive growth score model, F , integrating morphological and OAC indicators, was defined as the linear weighted sum of the retained k principal component scores. The weights depend on the explanatory power of each principal component regarding the total variance. The specific calculation formulas are as follows:

$$F = \sum_{j=1}^k w_j \cdot P_j, \quad (2)$$

$$w_j = \frac{\theta_j}{\sum_{l=1}^k \theta_l}, \quad (3)$$

where P_j is the score of a specific principal component, θ_j is the variance contribution rate of that principal component, and w_j is the normalized weight coefficient.

Considering the inherent heterogeneity in the initial volume and state of organoids upon seeding, a static score at a single time point struggles to accurately reflect the impact of the drug on the organoids. Therefore, this study introduced a self-control strategy, defining the relative growth score, ΔF , as the final evaluation metric for drug

response:

$$\Delta F = F_{after} - F_{before}, \quad (4)$$

This metric quantifies the change in the growth state of the organoid population between two time points (achieving the third level of dimensionality reduction). A larger ΔF indicates a better relative growth level of the organoid population.

2.7. Statistical analysis

The K-means ++ and PCA algorithms were implemented based on the scikit-learn machine learning library in python. The Pearson correlation coefficient was employed to analyze the consistency between ΔF and the gold-standard ATP data. Comparisons among multiple groups were performed using one-way analysis of variance. The Wilcoxon's t-test was employed to assess the statistical significance of the difference between the dependent correlation coefficients derived from the two models evaluated on the same batch of samples. $P < 0.05$ was considered statistically significant.

3. Results

3.1. Dynamic evolution of organoid OAC under drug treatment

Fig. 3 illustrates the three-dimensional rendering results of the organoid OAC for groups treated with varying drug concentrations before drug administration (Day 3) and 48 hours post-treatment (Day 5). Prior to drug administration (Day 3), the mean attenuation coefficients of the organoids in wells C3 (20 μM), B6 (40 μM), and B9 (80 μM) were 1.553 mm^{-1} , 1.526 mm^{-1} , and 1.544 mm^{-1} , respectively. In the 3D OAC renderings, the organoids predominantly exhibited blue/green hues, particularly the large-sized organoids (indicated by the arrows in Fig. 3(a1-d1)).

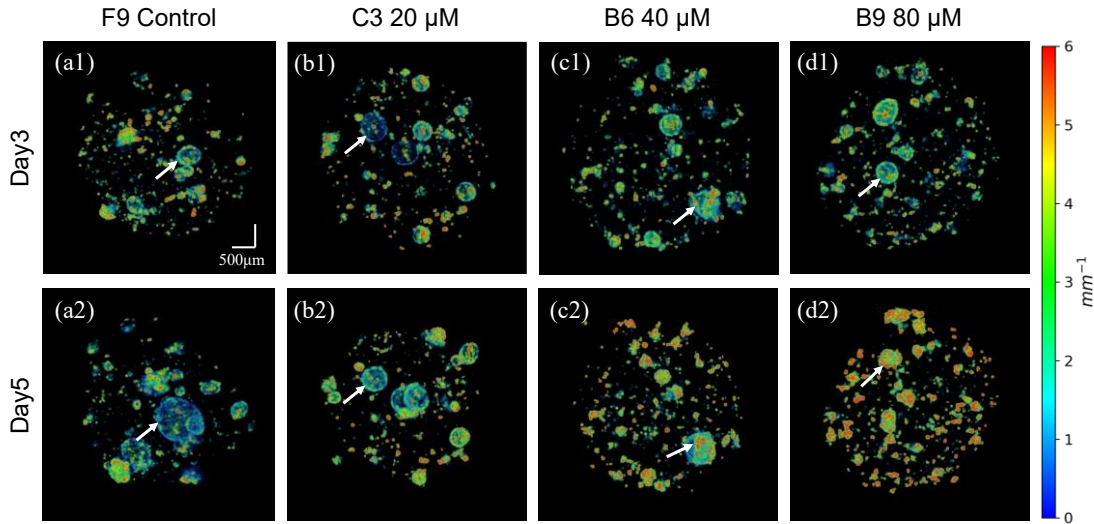


Fig. 3. Evolution of the three-dimensional distribution of OAC in organoids before and after drug treatment. (a-d) Represent the blank control group and the icaritin treatment groups at concentrations of 20, 40, and 80 μM , respectively. The top and bottom rows display 3D volumetric renderings of OAC at Day 3 (pre-treatment) and Day 5 (48 h post-treatment). The color bar on the right indicates OAC values (mm^{-1}), with blue/green hues representing low OAC and yellow/red hues representing high OAC. White arrows highlight changes in OAC signal intensity within large-sized organoids under different treatment conditions.

Following drug treatment, the mean OACs of the organoids in the treated wells C3, B6, and B9 increased to 1.591 mm^{-1} , 1.672 mm^{-1} , and 1.830 mm^{-1} , respectively. The magnitude of this increase demonstrated a positive correlation with the drug concentration (representing increases of 2.45%, 9.57%, and 18.52%, respectively). In the corresponding OAC renderings, the organoids transitioned to predominantly yellow/red hues, which was especially evident in the large-sized organoids (indicated by the white arrows in Fig. 3(c2) and (d2)). Unlike the drug-treated wells, the organoids in the F9 control well maintained their predominantly blue/green hues, and their mean OAC decreased from 1.539 mm^{-1} to 1.308 mm^{-1} , representing a reduction of 15.01%. Notably, the mean OAC of the specific organoid indicated by the arrows in Fig. 3(a1) and (a2) showed a substantial reduction from 1.408 mm^{-1} to 0.967 mm^{-1} , a decrease of 31.32%.

Furthermore, a comparative analysis was performed on the relative change rates and relative response ratios of the fill volume and OAC of the organoid populations across all 24 experimental wells, with the statistical results presented in Fig. 4(A). Regarding fill volume, the control organoid population exhibited a significant growth state; their mean fill volume markedly expanded from $(3.668 \pm 1.625) \times 10^{-4} \text{ mm}^3$ on Day 3 to $(13.96 \pm 4.30) \times 10^{-4} \text{ mm}^3$ on Day 5, achieving a relative growth rate of 223.8%. In contrast, growth in the drug-treated groups was evidently inhibited, with the degree of inhibition intensifying as the drug concentration increased. Specifically, the mean fill volume of the $20 \mu\text{M}$ group increased from $(3.572 \pm 0.554) \times 10^{-4} \text{ mm}^3$ to $(5.474 \pm 1.257) \times 10^{-4} \text{ mm}^3$ (a growth rate of 57.0%). The $40 \mu\text{M}$ group exhibited a marginal increase from $(3.324 \pm 0.544) \times 10^{-4} \text{ mm}^3$ to $(3.600 \pm 0.630) \times 10^{-4} \text{ mm}^3$ (an 8.3% growth rate), whereas the $80 \mu\text{M}$ high-concentration group experienced a reduction from $(4.584 \pm 2.326) \times 10^{-4} \text{ mm}^3$ to $(3.407 \pm 0.957) \times 10^{-4} \text{ mm}^3$, representing a negative growth rate of -16.2%.

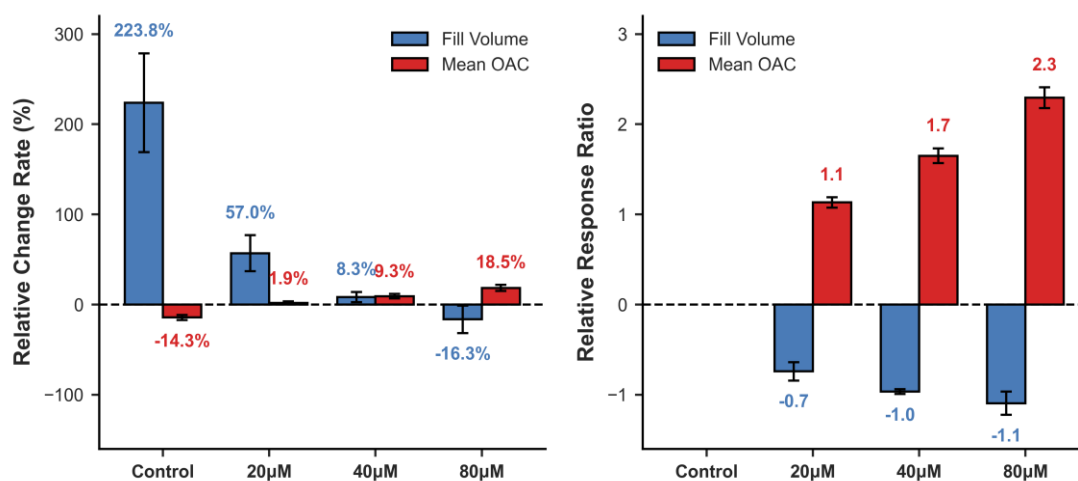


Fig. 4. Relative change rates and response ratios of mean fill volume and mean OAC in organoid populations under icaritin treatment. (A) Relative change rates from Day 3 to Day 5 across different concentration groups, calculated as $(\text{Day 5} - \text{Day 3}) / \text{Day 3}$. (B) Relative response ratios of drug-treated groups compared to the blank control, calculated as $(\Delta_{\text{Drug}} - \Delta_{\text{Ctrl}}) / |\Delta_{\text{Ctrl}}|$, where Δ represents the change from Day 3 to Day 5 for each group. Blue and red bars represent the mean fill volume and mean OAC, respectively.

Contrary to the volumetric changes, the OAC demonstrated a significant dose-dependent elevation. The mean OAC of the control organoid population decreased from $1.36 \pm 0.22 \text{ mm}^{-1}$ on Day 3 to $1.15 \pm 0.19 \text{ mm}^{-1}$ on Day 5, representing a relative reduction of 14.3%. Conversely, in the drug-treated groups, the OAC exhibited an upward trend with increasing drug concentrations. Specifically, the $20 \text{ }\mu\text{M}$ group rose from $1.73 \pm 0.25 \text{ mm}^{-1}$ to $1.76 \pm 0.28 \text{ mm}^{-1}$ (a 1.9% relative increase); the $40 \text{ }\mu\text{M}$ group increased from $1.97 \pm 0.31 \text{ mm}^{-1}$ to $2.15 \pm 0.33 \text{ mm}^{-1}$ (a 9.3% increase); and the $80 \text{ }\mu\text{M}$ group further increased from $1.65 \pm 0.29 \text{ mm}^{-1}$ to $1.92 \pm 0.35 \text{ mm}^{-1}$, reaching an 18.5% increase. Further calculation of the relative response ratio revealed, as shown in Figure 4(B), that the response intensity of OAC was superior to that of volume across all evaluated concentrations. Notably, at the $80 \text{ }\mu\text{M}$ concentration, the relative response ratio of OAC reached 2.29, which was significantly higher than the response level of volume (absolute value of 1.09). These results demonstrate that, compared to volumetric changes, OAC can more accurately capture early tissue alterations, exhibiting potential as a highly sensitive evaluation metric for drug response.

3.2. Type heterogeneity and drug response analysis based on multi-dimensional features

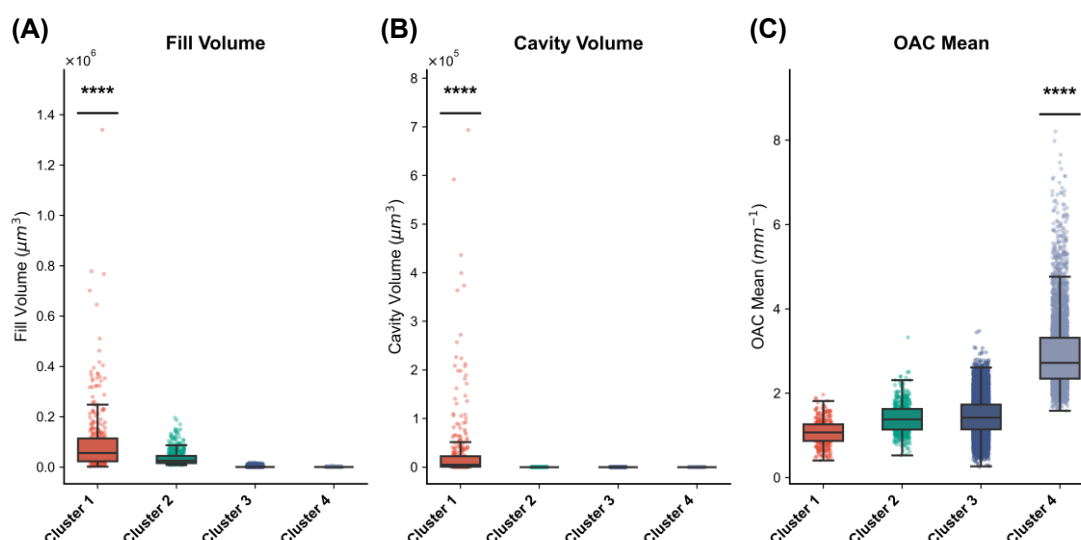


Fig. 5. Typical feature distributions of organoid classes identified by clustering. (A-C) Boxplots illustrating the distribution differences among the four organoid clusters (Cluster 1-4) in Fill Volume, Cavity Volume, and Mean Optical Attenuation Coefficient, respectively. Scatter points represent individual organoid samples, and asterisks (****) indicate statistical significance ($P < 0.0001$).

Based on the aforementioned unsupervised clustering model integrating morphological parameters and the OAC, the organoids were clustered into four classes (Clusters 1-4). Statistical analysis of the 23,439 organoid samples from 24 wells across two time points used to construct the clustering model revealed that cavity structure, fill volume, and mean OAC were the key features distinguishing the different types. As shown in Fig. 5, Cluster 1 was the only population among all types exhibiting a cystic structure. Clusters 2, 3, and 4 constituted the solid organoid cluster; their common feature was a cavity volume of 0, presenting a completely solid structure. Specifically,

Cluster 2 was characterized as solid, large-volume, and low-OAC; it was the second-largest volume type after Cluster 1 and maintained a relatively low OAC. Cluster 3 was characterized as solid, small-volume, and low-OAC. Cluster 4 represented the opposite extreme, characterized as solid, extremely small-volume, and high-OAC.

Table 1. Statistical analysis results of feature parameters based on clustering (Well B9).

Stage	Cluster	Number	Fill Volume average ($\times 10^{-4} \text{ mm}^3$)	Surface Area average (mm^2)	Cavity Volume average ($\times 10^{-3} \text{ mm}^3$)	Cyst Thickness average (mm)
Before	1	7	94.37 $\pm$ 53.31	1.83 $\pm$ 1.04	2.02 $\pm$ 2.86	0.010 $\pm$ 0.003
	2	12	31.90 $\pm$ 20.82	0.62 $\pm$ 0.40	0	0
	3	317	1.01 $\pm$ 1.98	0.02 $\pm$ 0.04	0	0
	4	104	0.36 $\pm$ 0.36	0.01 $\pm$ 0.01	0	0
After	1	0	0	0	0	0
	2	14	52.30 $\pm$ 44.10	1.02 $\pm$ 0.86	0	0
	3	272	1.37 $\pm$ 2.82	0.03 $\pm$ 0.05	0	0
	4	101	0.50 $\pm$ 0.57	0.01 $\pm$ 0.01	0	0

Stage	Cluster	Sphericity average (arb.units)	Longest Axis length average (mm)	Shortest Axis length average (mm)	OAC Mean average (mm^{-1})	OAC Std average (mm^{-1})
Before	1	0.25 $\pm$ 0.06	0.737 $\pm$ 0.268	0.282 $\pm$ 0.094	1.385 $\pm$ 0.265	1.329 $\pm$ 0.219
	2	0.28 $\pm$ 0.05	0.513 $\pm$ 0.164	0.178 $\pm$ 0.069	1.444 $\pm$ 0.211	1.402 $\pm$ 0.201
	3	0.82 $\pm$ 0.28	0.101 $\pm$ 0.062	0.033 $\pm$ 0.028	1.562 $\pm$ 0.351	1.139 $\pm$ 0.400
	4	0.87 $\pm$ 0.21	0.078 $\pm$ 0.028	0.026 $\pm$ 0.011	2.777 $\pm$ 0.806	2.106 $\pm$ 0.496
After	1	0	0	0	0	0
	2	0.27 $\pm$ 0.04	0.665 $\pm$ 0.303	0.221 $\pm$ 0.101	1.798 $\pm$ 0.331	1.518 $\pm$ 0.318
	3	0.81 $\pm$ 0.30	0.117 $\pm$ 0.082	0.039 $\pm$ 0.035	1.824 $\pm$ 0.415	1.250 $\pm$ 0.409
	4	0.82 $\pm$ 0.23	0.091 $\pm$ 0.041	0.029 $\pm$ 0.014	3.537 $\pm$ 1.253	2.285 $\pm$ 0.485

Based on the clustering results, statistical analysis corresponding to organoid classes was performed for individual organoid populations. Table 1 presents the statistical results for well B9 before and after drug administration, including the count of each class and their mean morphological parameters and OACs. Cluster 1 possessed the largest mean fill volume ((94.37 $\pm$ 53.31) $\times 10^{-4} \text{ mm}^3$) and contained significant internal cavities (mean cavity volume of (2.02 $\pm$ 2.86) $\times 10^{-3} \text{ mm}^3$), with the cavity volume accounting for approximately 21.5%. In contrast to its massive apparent volume, its mean OAC was the lowest among all types (1.385 $\pm$ 0.265 mm^{-1}). The mean fill volume of Cluster 2 reached (31.90 $\pm$ 20.82) $\times 10^{-4} \text{ mm}^3$, and it maintained a relatively low OAC (1.444 $\pm$ 0.211 mm^{-1}), which was only slightly higher than that of Cluster 1. The mean fill volume of Cluster 3 ((1.01 $\pm$ 1.98) $\times 10^{-4} \text{ mm}^3$) was significantly smaller than that of the first two classes, while its OAC (1.562 $\pm$ 0.351 mm^{-1}) was slightly higher than those of Clusters 1 and 2, though it remained within a relatively low range without drastic signal enhancement. Cluster 4 had the smallest mean fill volume at only (0.36 $\pm$ 0.36) $\times 10^{-4} \text{ mm}^3$, but its mean OAC notably increased to 2.777 $\pm$ 0.806 mm^{-1} , which was

approximately twice that of Cluster 1.

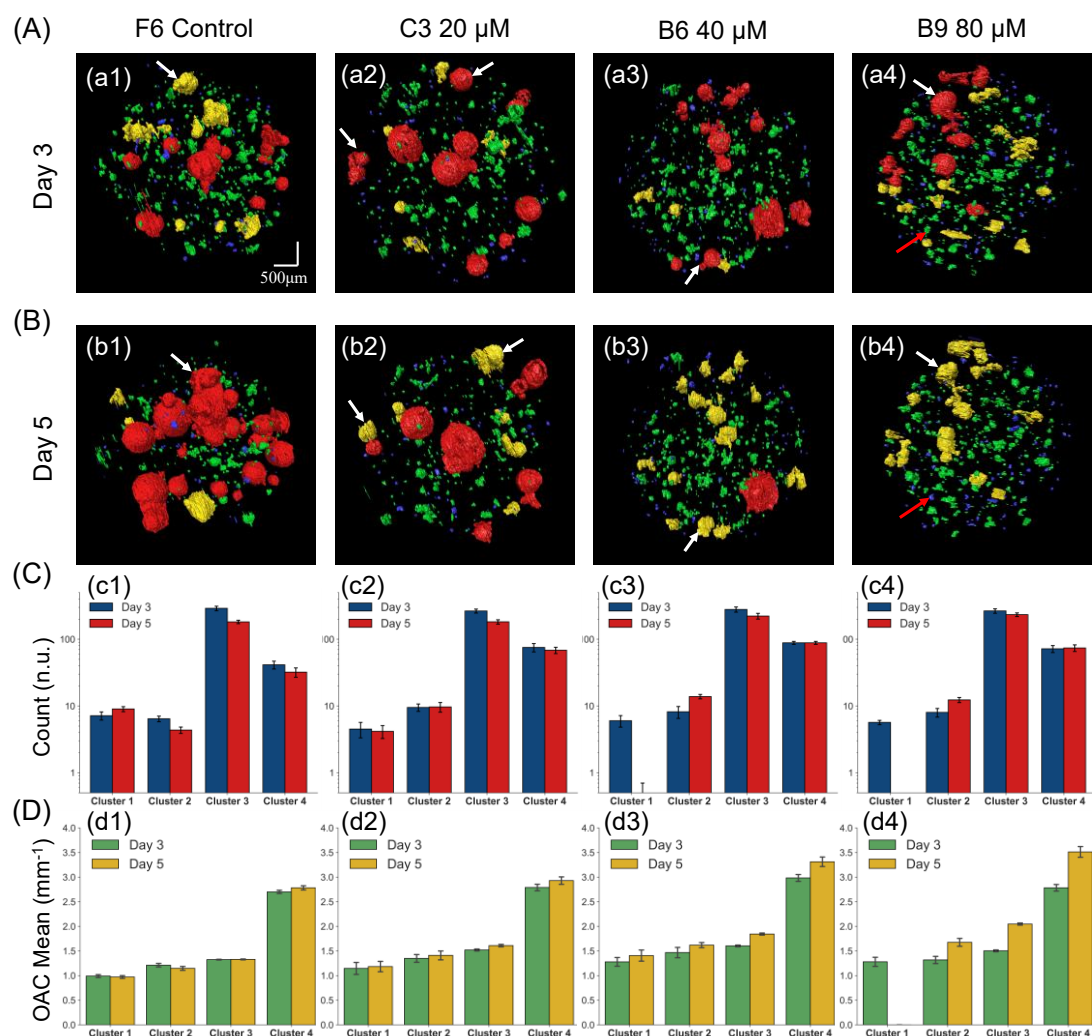


Fig. 6. Evolution of organoid characteristics under icaritin treatment. (A, B) 3D pseudo-color clustering renderings of the blank control group and icaritin treatment groups with different concentrations (20, 40, 80 μM) at Day 3 (pre-treatment) and Day 5 (post-treatment). Colors correspond to distinct cluster categories: Red (Cluster 1, large cystic), Yellow (Cluster 2, large solid), Green (Cluster 3, small solid), and Blue (Cluster 4, high-density small solid). White arrows indicate the transformation process of specific large organoids from cystic (Cluster 1) to solid (Cluster 2), while red arrows indicate the transformation of specific organoids from Cluster 3 to Cluster 4. (C) Statistical changes in the counts of the four organoid classes across experimental groups. (D) Statistical changes in the mean Optical Attenuation Coefficient of the four organoid classes across experimental groups.

Utilizing the category information obtained from clustering, class visualization was performed for the organoids within each population, where red, yellow, green, and blue hues correspond to Clusters 1-4, respectively. Fig. 6(A-B) displays the class visualization results for the control and drug-treated groups before and after drug administration. Fig. 6(C) illustrates the changes in the count of different organoid classes within each population before and after treatment. In the control group, the count of Cluster 1 increased from 8 (Day 3) to 13 (Day 5), and the count of Cluster 2 decreased from 9 (Day 3) to 6 (Day 5), with instances of Cluster 2 on Day 3 transforming into Cluster 1 on Day 5 (indicated by the white arrows in well F9). In the drug-treated groups,

as the drug concentration increased (20, 40, and 80 μM), the count of Cluster 1 gradually decreased and completely disappeared in the 80 μM group. Conversely, the count of Cluster 2 increased with higher drug concentrations, with instances of Cluster 1 on Day 3 transforming into Cluster 2 on Day 5 (indicated by the white arrows in wells C3, B6, and B9). In both the control and drug-treated groups, the count of Cluster 3 exhibited a decreasing trend, although this reduction slowed down as the drug concentration increased. The situation for Cluster 4 was similar to that of Cluster 3; however, at the 80 μM concentration, a slight increase in count occurred, and transitions from Cluster 3 on Day 3 to Cluster 4 on Day 5 were observed (indicated by the red arrow in well B9).

Fig. 6(D) presents the changes in the mean OAC of different organoid classes within each population before and after treatment. Compared to pre-treatment levels, the mean OAC of Cluster 1 in the treated groups increased at 20 and 40 μM . However, at 80 μM , because the Cluster 1 organoids transformed into Cluster 2 post-treatment, the resulting mean OAC for Cluster 1 was recorded as 0. In the blank control group, the mean OACs of Clusters 1 and 2 on Day 5 slightly decreased compared to Day 3. The mean OAC of Cluster 4 showed a minor increase, but the magnitude was notably smaller than that in the drug-treated groups. These results indicate that under the action of icaritin, there is, on one hand, a volume reduction and complete loss of the characteristic cavity structure in Cluster 1, and on the other hand, a relative increase in dense organoids. Consequently, this manifests as a positive correlation between the mean OAC of the organoid cluster and the drug concentration.

3.3. Construction and validation of the comprehensive growth score model

Based on the procedures described in Section 2.6, we constructed the comprehensive growth score model for the organoids. The analysis results showed that the eigenvalues of the first four principal components (PCA1-PCA4) were all greater than 1, with a cumulative variance contribution rate reaching 70.82%, indicating that this model effectively retained the vast majority of the morphological and OAC information. Among them, the first principal component (PCA1) had the highest variance contribution rate (33.16%) and served as the dominant factor determining the model's evaluation performance.

The rotated loading matrix is shown in Table 2, where X_{ab} represents the b -th feature of the a -th class. Further analysis revealed that PCA1, as the factor explaining the largest variance, exhibited positive loadings on X_{11} (Cavity Volume₁, +0.2902) and X_{12} (Surface Area₁, +0.2829); however, its weights in the negative direction were more prominent and concentrated. Specifically, X_{31} (Cyst Thickness₃) possessed the largest absolute negative loading value, reaching -0.3636, followed by shortest-axis related indicators such as X_{34} (Shortest Axis₃, -0.3524) and X_{43} (Shortest Axis₄, -0.3227). Additionally, the quantity indicator X_{21} (Number₂, -0.3482) also demonstrated a negative correlation, indicating that mathematically, PCA1 primarily captured the relationships among volume, surface area, cyst thickness, axial dimensions, and cluster quantity. The high loadings of PCA2 were mainly concentrated in Cluster 2 with a highly consistent

direction. Among them, the loading value of X_{23} (Surface Area₂) reached +0.5646, making it the variable with the largest contribution to this principal component, closely followed by X_{24} (Fill Volume₂, +0.5250) and X_{22} (Shortest Axis₂, +0.4919). In contrast, the loadings of PCA2 on other variables were generally low (mostly within ± 0.2), indicating that this principal component primarily explained the morphological variation within Cluster 2. PCA3 mainly reflected the internal differences of Cluster 3; on one hand, X_{32} (Number₃) exhibited an extremely strong negative loading (-0.6079), which was the item with the largest absolute value in that column; on the other hand, X_{35} (Surface Area₃) exhibited a strong positive loading (+0.4913). In the PCA4 principal component, the roughness indicator X_{13} (Roughness₁) had one of the highest negative loadings in the entire table (-0.5656), while the longest-axis indicator X_{41} (Longest Axis₄) demonstrated a significant positive loading (+0.5063).

Table 2. Loading Matrix of Principal Components.

Feature Name	PCA1	PCA2	PCA3	PCA4
Cavity Volume ₁ (X_{11})	0.2902	0.0716	0.1370	-0.2865
Surface Area ₁ (X_{12})	0.2829	0.1951	0.1218	-0.2574
Roughness ₁ (X_{13})	0.0039	-0.2391	-0.3460	-0.5656
Number ₂ (X_{21})	-0.3482	-0.1525	-0.1542	-0.2212
Shortest Axis ₂ (X_{22})	-0.2063	0.4919	-0.1419	-0.0575
Surface Area ₂ (X_{23})	-0.1171	0.5646	-0.0287	-0.0775
Fill Volume ₂ (X_{24})	-0.1531	0.5250	0.0109	-0.1794
Cyst Thickness ₃ (X_{31})	-0.3636	-0.0819	0.1040	-0.1573
Number ₃ (X_{32})	-0.0971	-0.0549	-0.6079	-0.1238
OAC Mean ₃ (X_{33})	-0.2456	-0.0231	0.0916	-0.1021
Shortest Axis ₃ (X_{34})	-0.3524	-0.0669	0.2623	-0.3008
Surface Area ₃ (X_{35})	-0.2377	-0.1042	0.4913	-0.0754
Longest Axis ₄ (X_{41})	-0.2732	0.0399	-0.0897	0.5063
OAC Mean ₄ (X_{42})	-0.2710	-0.1058	0.1868	-0.0509
Shortest Axis ₄ (X_{43})	-0.3227	-0.0401	-0.2450	0.1890

Based on the variance contribution rate of each principal component, the normalized weight coefficients were calculated as $w_1=0.468$, $w_2=0.247$, $w_3=0.165$, and $w_4=0.120$, respectively. Accordingly, the specific comprehensive growth score model was constructed as shown in Equation (5). By multiplying the loading of each principal component by its variance weight and summing the results, the final contribution weight of each original feature in the comprehensive growth score model could be calculated. For example, the comprehensive weight of the OAC feature X_{42} was -0.128, presenting a significant negative contribution.

$$F = 0.468 \cdot P_1 + 0.247 \cdot P_2 + 0.165 \cdot P_3 + 0.120 \cdot P_4 \quad (5)$$

Linear regression analysis was performed between the model calculation results and the ATP bioluminescence viability data to validate the accuracy of the comprehensive growth score model in quantifying drug efficacy. The validation process was divided into endpoint assessment and relative assessment.

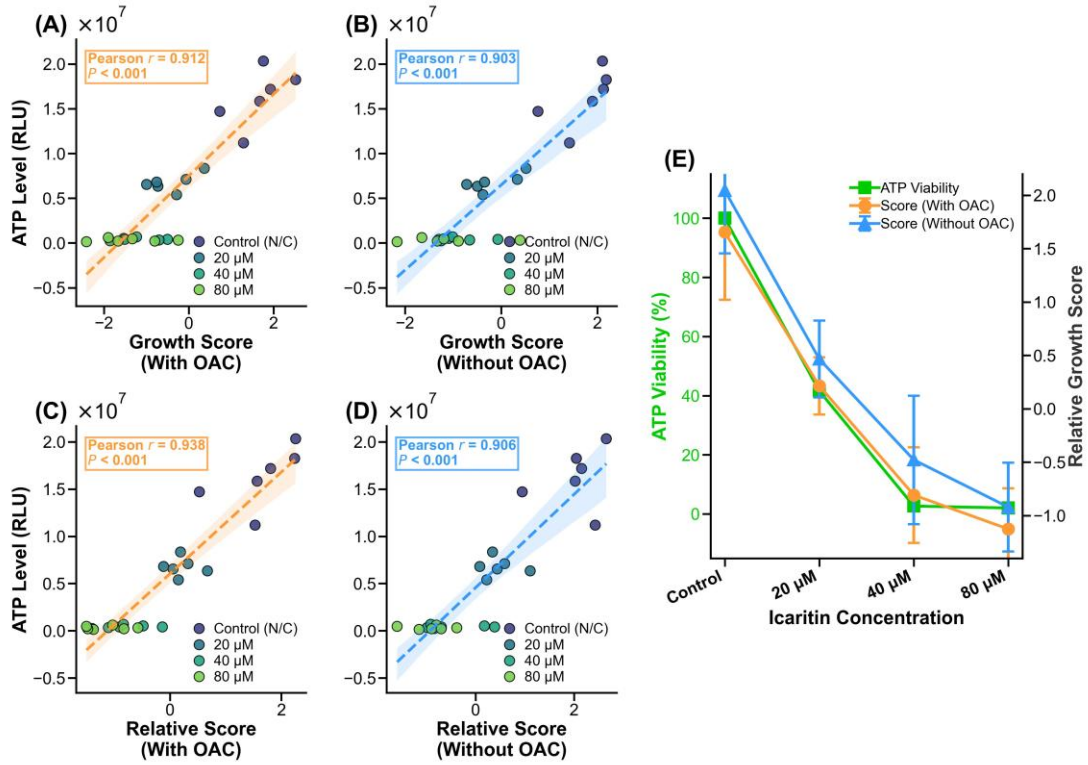


Fig. 7. Linear regression results of endpoint and relative growth scores against ATP levels for two models. (A, B) Linear regression analysis of the endpoint growth score at Day 5 versus ATP levels for the model fusing OAC with multi-dimensional morphological features ("with OAC") and the morphology-only model ("without OAC"), respectively. (C, D) Linear regression analysis of the relative growth score versus ATP levels for the two models mentioned above. (E) Trend comparison of ATP relative viability (green line) and relative growth scores from both models (orange and blue lines) across different drug concentrations.

First, the correlation regression analysis results for the experimental endpoint (Day 5) scores and ATP levels (Fig. 7(A-B)) showed that the endpoint scores of both the model fusing OAC with multi-dimensional morphological features and the morphology-only model were significantly positively correlated with ATP viability ($P < 0.001$). Notably, the fusion model exhibited a higher linear fit, with a Pearson correlation coefficient of $r = 0.912$, which outperformed the morphology-only model ($r = 0.903$). This result indicates that at a single time point, introducing the attenuation coefficient reflecting internal density can more accurately represent the growth state of organoids than purely morphological features.

Further validation of the relative growth score following the introduction of the temporal dimension (Fig. 7(C-E)) revealed that the advantages of the model were further amplified after implementing the self-control strategy. The correlation coefficient between the ΔF of the fusion model and ATP increased to $r = 0.938$ ($P < 0.001$), achieving exceptionally high consistency. After removing the attenuation coefficient features, the correlation coefficient between ΔF and ATP for the morphology-only model was only $r = 0.906$, which was lower than that of the fusion model. Statistical comparison using the Williams' t-test confirmed that the correlation improvement brought by integrating

OAC features is statistically significant ($P=0.0459$). These results indicate that using the relative growth score to quantify the organoid growth level can effectively eliminate initial interference; its biological rationale lies in the fact that the measured ATP value essentially reflects the survival quality of the organoids rather than their physical scale alone.

Table 3. Growth level assessment results of organoid populations in icaritin-treated groups used for modeling.

No.	Drug Concentration (μM)	Wells	Growth Score		Relative	ATP
					Growth Score	(RLU)
			Day 3	Day 5	Day 5-Day 3	Day 5
1	Control	E11	0.205	0.735	+0.530	14700000
2		F2	0.286	2.524	+2.238	18240000
3		F6	-0.493	1.763	+2.256	20320000
4		F8	0.122	1.929	+1.807	17170000
5		F9	0.111	1.677	+1.566	15830000
6		F11	-0.236	1.293	+1.529	11180000
7	20	B2	-0.439	-0.286	+0.153	5391000
8		B3	-1.051	-0.993	+0.058	6538000
9		B4	-0.388	-0.062	+0.326	7103000
10		C2	-1.400	-0.726	+0.674	6336000
11		C3	0.178	0.369	+0.191	8336000
12		C4	-0.643	-0.756	-0.113	6800000
13	40	B5	-1.055	-1.533	-0.478	511900
14		B6	-0.365	-0.500	-0.135	404500
15		B7	-0.605	-1.509	-0.904	403900
16		C5	0.404	-0.699	-1.103	330900
17		C6	-0.458	-1.855	-1.397	238900
18		C7	-0.398	-1.23	-0.832	682100
19	80	B8	0.326	-0.246	-0.572	312700
20		B9	-1.042	-2.412	-1.370	140300
21		B10	-0.872	-1.897	-1.025	601300
22		C8	-0.193	-1.666	-1.473	211300
23		C9	0.166	-1.330	-1.496	465900
24		C10	0.024	-0.800	-0.824	211800

The evaluation results of the organoid cluster growth levels in the experimental groups are presented in Table 3. In the 80 μM high-concentration icaritin treatment group (e.g., Well C10), the organoids exhibited extremely low ATP viability (2.12×10^5 RLU), indicating a state of severe apoptosis; at this stage, the relative growth score calculated by the model was a substantial negative value ($\Delta F = -0.824$). A deep breakdown of its score contributions revealed that the dominant factor driving this sharp decline was not simply volume reduction, but rather a significant increase in the attenuation coefficient (corresponding to feature X_{42}) and a sharp surge in surface roughness

(X_{13}) at Day 5. This confirms that by capturing changes in optical attenuation characteristics, the model can identify samples that maintain their apparent morphology but experience a shift in health status, accurately reflecting the growth inhibitory effects under high drug concentrations.

Conversely, in the 20 μM low-concentration group (e.g., Well B4), despite slight drug interference, the organoids maintained high ATP levels (7.10×10^6 RLU), and the model yielded a positive ΔF (+0.326). This indicates that at this concentration, the positive contribution of morphological feature changes to the model outweighed the alterations in the attenuation coefficient. This demonstrates that the model possesses excellent specificity, capable of distinguishing between growth inhibition and proliferation states, thereby effectively avoiding the risk of misjudging low-toxicity responses as strong positives. More importantly, this model is more accurate than the traditional ATP population-averaging method, acutely capturing differential effects within the same concentration group. In the 40 μM medium-concentration group, although Wells C6 and C7 received identical drug treatments, their ΔF values were -1.397 and -0.832, respectively; this difference was highly consistent with the ATP detection results (2.39×10^5 vs. 6.82×10^5 RLU). This further confirms that compared to ATP population-averaged readings that obscure individual differences, the organoid growth level evaluation framework proposed in this study preserves the true characteristics of the samples and provides a more accurate analytical tool for evaluating drug efficacy.

3.4. Validation of the organoid cluster growth level model based on combinational drug regimens

To verify the universality and robustness of the organoid comprehensive growth score model—integrating OAC and multi-dimensional morphological features—constructed in this study across different drug mechanisms and combination therapies, first-line chemotherapy drugs for hepatobiliary tumors, including 5-Fluorouracil (5-FU), Gemcitabine, Cisplatin, and Oxaliplatin, were selected. Six experimental regimens, including monotherapy and dual-drug combinations, were designed (as shown in Table 4).

Fig. 8 visually presents the effects of different chemotherapy regimens on the evolution of organoid types. By comparing the clustering distribution changes between Day 3 (pre-treatment) and Day 5 (48 hours post-treatment), distinct drug effects can be observed: In the blank control group, such as wells F6 and F8, the organoids exhibited a mixed distribution of red, yellow, and green hues on Day 3. After 2 Days of culture (Day 5), the images were dominated by large red areas (Cluster 1), accompanied by a small amount of yellow (Cluster 2). This indicates that without drug interference, organoids tend to evolve towards larger volumes and more mature cystic cavities. Unlike the control group, the 5-FU treatment group (e.g., well C3) did not show a massive increase in the red type on Day 5; although a considerable proportion of red (Cluster 1) and yellow (Cluster 2) individuals remained in the field of view and the overall structure was kept intact, their volumetric expansion was evidently restricted. In the combinational chemotherapy groups containing Gemcitabine (E4: Gemcitabine + Oxaliplatin;

C11: Gemcitabine + Cisplatin), the large cystic red type (Cluster 1) present on Day 3 almost completely disappeared after drug administration, replaced by a large number of green (Cluster 3) and blue (Cluster 4) organoids. The disappearance of the large cystic type and the enrichment of small-volume types directly reveal the cytotoxic effects induced by the combinational drug treatment. By tracking the growth and decline of different organoid types, the clustering model successfully reflected drug effects at the single-organoid level.

Table 4. The combination therapy experimental regimen from another patient for validation.

No.	Drug Name	Wells	Drug Concentration
1	Control	F7、F8、F9	/
		F10、F11、F12	
2	5-FU	C3、C4、C5	15 µg/mL
		C6、C7、C8	
3	Gemcitabine + 5-FU	C7、C8、C9	25.95 µg/mL+15 µg/mL
		C10、C11、C12	
4	Gemcitabine + Cisplatin	D1、D2、D3	25.95 µg/mL+80 µg/mL
		D4、D5、D6	
5	Gemcitabine + Oxaliplatin	E1、E2、E3	25.95 µg/mL+0.814 µg/mL
		E4、E5、E6	
6	Oxaliplatin + 5-FU	F1、F2、F3	15 µg/mL+0.814 µg/mL
		F4、F5、F6	

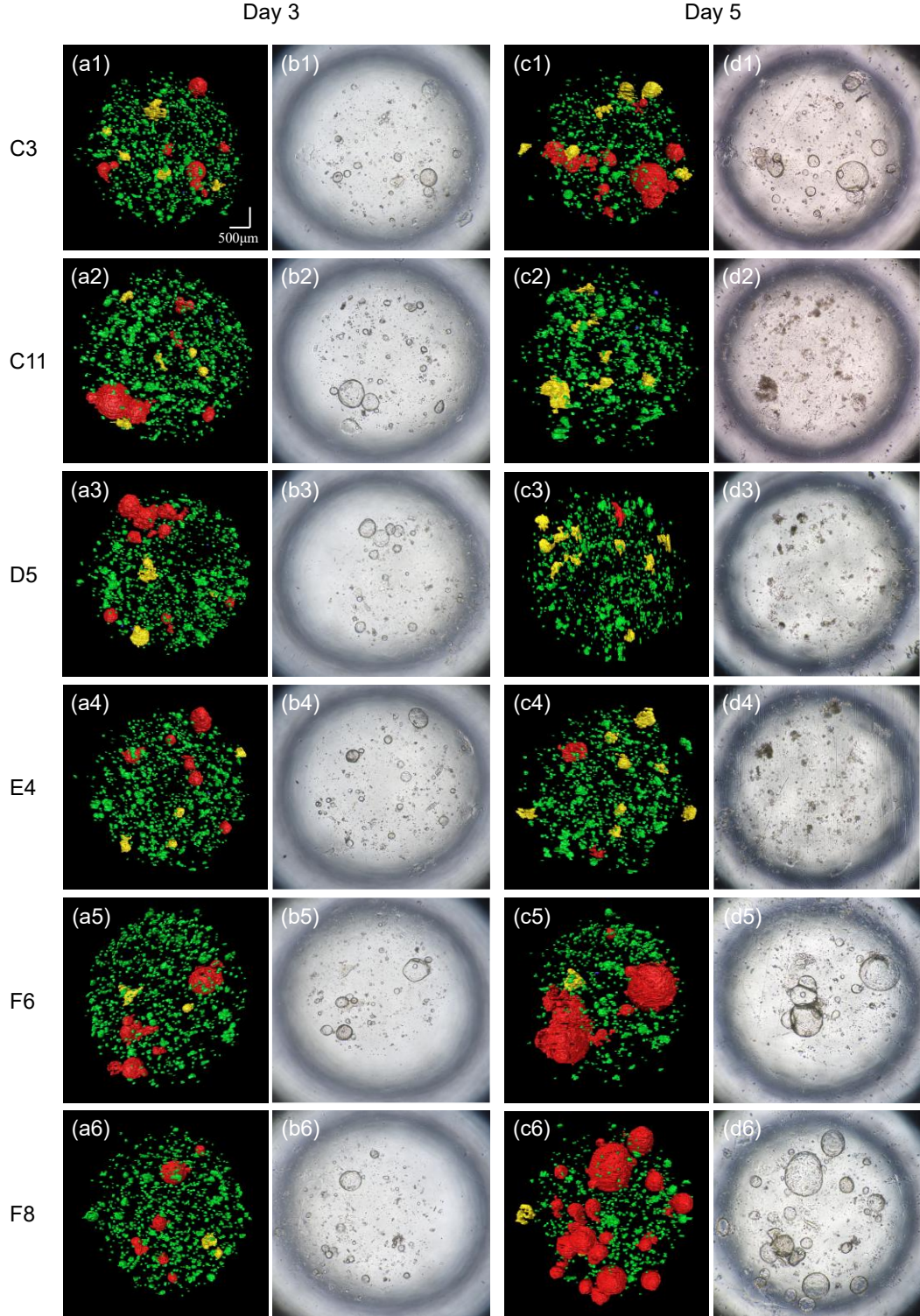


Fig. 8. Evolution of organoid phenotypes under combination drug treatments. (a1-a6) 3D OCT clustering renderings on Day 3; (b1-b6) brightfield microscopy images on Day 3; (c1-c6) 3D OCT clustering renderings on Day 5; (d1-d6) brightfield microscopy images on Day 5. Each row represents a different drug treatment regimen (see Table 4 for details).

Finally, the comprehensive growth score model integrating OAC and multi-dimensional morphological features was utilized to calculate the relative growth score for each group, which was then subjected to regression analysis with the ATP

bioluminescence assay results. As shown in Fig. 9(A), even under complex conditions encompassing multiple drug mechanisms, the model's relative growth score maintained a significant positive correlation with ATP levels ($r=0.887$, $P<0.001$). Further analysis of the comparison results in Fig. 9(B) revealed that although the relative growth score of the 5-FU group was lower than that of the control group, it remained positive. This is consistent with the retention of Cluster 1 and Cluster 2 in the clustering results, indicating that the model accurately determined that the organoids in this group were viable but subject to drug-induced growth inhibition. Conversely, the score for the Gemcitabine combination group was significantly negative. This corresponds to the marked reduction of Cluster 1 and Cluster 2, and the significant increase in Cluster 3 and Cluster 4 in the clustering results. In summary, the fusion model of OAC and multi-dimensional morphological features proposed in this study can distinguish organoid feature alterations caused by different chemotherapy regimens and achieves non-destructive drug efficacy evaluation highly consistent with the ATP gold standard.

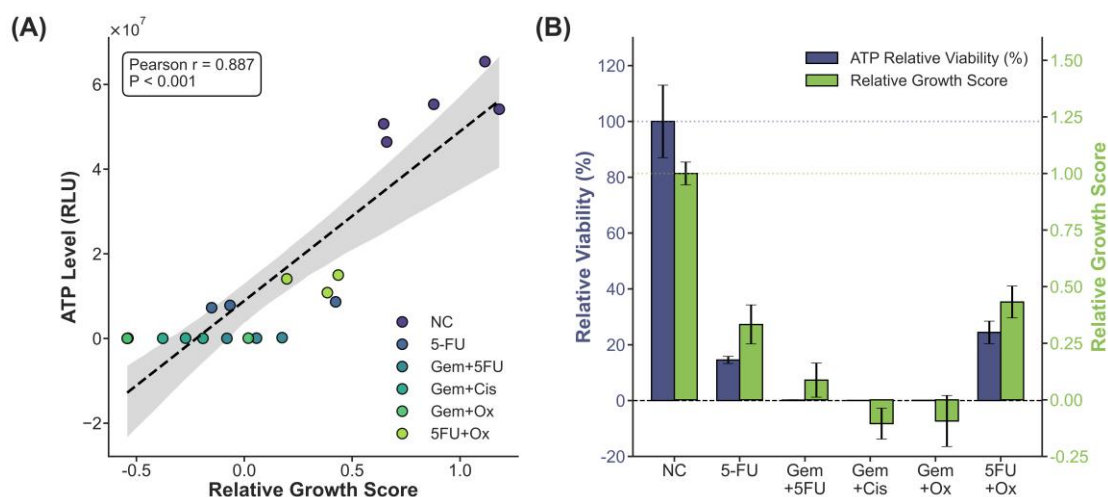


Fig. 9. Model validation and comparison under combination chemotherapy regimens. (A) Linear regression analysis of the relative growth score versus ATP levels across various chemotherapy regimens, including monotherapy (5-FU) and dual-drug combinations (e.g., Gem + Cis, Gem + Ox). Scatter points represent different experimental groups, the black dashed line indicates the fitted regression line, and the gray area represents the confidence interval. (B) Comparison of consistency between ATP relative viability (blue bars, left axis) and relative growth score (green bars, right axis) across different drug treatment groups.

4. Discussion

This study proposed and validated a non-destructive drug response evaluation framework for tumor organoids fusing the mean OAC and multi-dimensional morphological features, demonstrating that OCT-derived OAC can serve as a highly sensitive endogenous indicator to enhance the sensitivity of drug response evaluation models. Experimental data demonstrated that under icaritin treatment, the mean OAC of iCCA organoids exhibited a significant dose-dependent increase, wherein the mean OAC of the 80 μ M group elevated by 32.8% compared to the blank control group. This alteration in optical properties is closely associated with microscopic pathological processes during drug-induced apoptosis. According to Mie scattering theory, the backscattering

intensity of biological tissues is determined by the internal refractive index distribution. In healthy cystic organoids, cells are orderly arranged and the internal cavities are filled with homogeneous low-refractive-index fluid, thereby collectively exhibiting a low OAC. However, during drug-induced apoptosis, events such as pyknosis, karyorrhexis, chromatin condensation, and organelle remodeling occur, which significantly increase the complexity of subcellular structures and refractive index variations. Consequently, the observed elevation in OAC directly reflects the macroscopic optical signal manifestations of these microscopic pathological alterations, providing a label-free detection indicator for early cellular damage.

The superiority of this evaluation framework is rooted in the systematic integration across the "imaging-segmentation-modeling" dimensions, progressively surmounting the bottlenecks of conventional techniques. Firstly, at the data acquisition level, it achieved a paradigm shift from destructive endpoint testing to longitudinal non-destructive monitoring. Traditional methods such as ATP assays are destructive, and their readouts are severely compromised by inherent variations in the initial number, size, and viability of organoids across different wells, resulting in poor comparability. Leveraging the label-free, three-dimensional imaging capabilities of OCT, continuous monitoring of the identical batch of organoids before and after drug administration can be accomplished, allowing for the evaluation of growth levels at different time points. Building upon this, by calculating the relative changes in growth levels based on their own baselines, the impact derived from initial inter-sample heterogeneity was effectively eliminated.

Secondly, at the information parsing level, it realized a transition from holistic population analysis to the multi-dimensional resolution of individual organoids. Utilizing the previously developed ParaSAM2CNN deep learning model, this study accurately identified individual organoids within complex culture environments. This enabled the analysis to transcend the crude paradigm of treating the entire culture well as a homogeneous entity, shifting instead towards the parallel measurement of hundreds of independent organoids. A high-dimensional feature set was extracted for each organoid, encompassing macroscopic morphological parameters such as volume, sphericity, and surface roughness, as well as the depth-resolved metric of OAC, which established the data foundation for unveiling the heterogeneous responses of organoid subpopulations.

Finally, at the information fusion and quantification level, an intelligent model translating high-dimensional features into comprehensive drug response evaluations was constructed. Addressing the massive quantity of individual organoids, unsupervised clustering was employed to conduct phenotypic classification, revealing their differential responses to drugs and intuitively demonstrating the dynamic transitions between different clusters (e.g., from the cystic Cluster 1 to the solid Cluster 2) under drug exposure. Furthermore, PCA was applied to achieve dimensionality reduction and fusion of the clustered statistical features, thereby constructing a comprehensive growth level model. By assigning differential loading weights to OAC and morphological

parameters, this model allocated a comprehensive weight coefficient of -0.128 to the OAC metric; this negative weight encapsulated a penalty for the covert damaged state characterized by morphological integrity yet internal densification. The incorporation of OAC elevated the correlation coefficient between the model score and the ATP gold standard from $r=0.906$ to $r=0.938$, achieving a non-destructive, quantitative, and highly sensitive evaluation of drug response.

Additionally, the universality of the aforementioned model was validated using chemotherapeutic agents with diverse mechanisms of action (including 5-Fu, Gemcitabine, and combinatorial regimens). The relative growth score derived from this model maintained high consistency with ATP cell viability ($r=0.887$). This demonstrated the value of fusing OAC with morphological parameters to construct a highly sensitive and broadly applicable non-destructive detection method. Such non-destructive, longitudinal, and high-information-density evaluation characteristics hold significant translational potential for enabling reliable personalized drug sensitivity predictions using scarce PDTO samples.

Despite the aforementioned advantages, this study still possesses certain limitations that require refinement in subsequent research. Firstly, the spatial resolution of the current OCT system restricted the direct observation of fine alterations in subcellular structures such as mitochondria. Future investigations could be coupled with technologies like super-resolution microscopy to establish more precise correlations between optical parameters and subcellular events. Secondly, although our feature set is relatively comprehensive, integrating more complex texture features or higher-order spectral features may contribute to further enhancing the detection sensitivity for extremely early apoptotic changes. Thirdly, although the ParaSAM2CNN model was optimized against OCT speckle noise, its segmentation accuracy remains challenged when processing severe organoid adhesion caused by high-density culture or severely fragmented morphologies induced by highly toxic drugs. Future studies could adopt more advanced instance segmentation networks or introduce topological constraint mechanisms to enhance the robustness of the algorithm. Finally, while our comprehensive growth score model demonstrated preliminary applicability across multiple chemotherapeutic regimens, the current independent validation was conducted on a limited number of iCCA patient-derived lines. Given the high inter-patient heterogeneity inherent to hepatobiliary tumors, future large-scale studies incorporating diverse patient-derived lines are essential to definitively validate the generalization capability of this evaluation framework.

In summary, this study developed a novel tumor organoid drug response evaluation framework based on the fusion of OCT-derived OAC and multi-dimensional morphological features. Through a coherent pipeline of continuous imaging, individual organoid segmentation, and feature fusion modeling, this framework systematically elevated the longitudinal comparative capability, heterogeneity resolution, and early detection sensitivity of the evaluation. We demonstrated that OAC is a sensitive indicator reflecting drug-induced cellular damage, and its fusion with morphological parameters

can quantify drug efficacy more comprehensively and earlier. This work provides a robust, quantitative, and non-destructive analytical tool for advancing the application of organoid models in precision medicine and drug discovery.

5. Conclusions

To address the key limitations of traditional tumor organoid drug response evaluation methods, including their destructive nature, reliance on endpoint testing, and susceptibility to inter-sample inconsistency, this study constructed a non-destructive evaluation framework based on OCT and machine learning. The results demonstrated that the OAC extracted from OCT signals can serve as a sensitive endogenous optical indicator, exhibiting a significant dose-dependent increase under icaritin treatment. This study systematically integrated the OAC metric, which reflects tissue density, with multi-dimensional morphological features for the first time, establishing a novel integrated evaluation framework of "imaging-segmentation-characterization-modeling". Based on clustering analysis and PCA, a comprehensive growth level model fusing OAC and multi-dimensional morphological parameters was constructed, achieving the non-destructive and quantitative evaluation of drug response. The correlation between the comprehensive growth level score obtained by this model and the ATP gold standard ($r=0.938$) was significantly superior to that of the morphology-only model. This framework overcame the difficulties that traditional endpoint methods cannot track longitudinally and that a single morphological parameter is insensitive to early internal densification damage. It provided a powerful non-destructive analytical tool for precision drug sensitivity testing based on PDOs, and demonstrated preliminary applicability across chemotherapy regimens with various mechanisms of action ($r=0.887$). In summary, this study developed a novel tumor organoid drug response evaluation method fusing OAC and multi-dimensional morphological information, providing an innovative technical solution with high sensitivity and potential generalizability to advance the practical application of organoid models in anti-tumor drug discovery and personalized medicine.

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Disclosures

The authors declare no conflicts of interest.

Data availability statement

Data underlying the results presented in this paper are not publicly available at this time due to intellectual property and pending patent applications, but may be obtained from the authors upon reasonable request.

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